F102 Study Guide — Topic Summaries

18 topics ordered by exam frequency

#1 Health and care products HIGH PRIORITY

36 exam questions | last tested 2025 | recency 15.27

What You Must Know

This topic covers the structure, features, risks, and capital requirements of health and care insurance products, focusing on Income Protection (IP), Critical Illness (CI), and Long-Term Care Insurance (LTCI). These products are frequently examined due to their complex benefit designs, unique morbidity risks, and the actuarial challenges in pricing, reserving, and managing them.

Core Concepts

Income Protection (IP) Contracts

- Provide regular income if the insured is incapacitated due to illness or accident, until recovery, death, or policy expiry.
- Benefits are typically a percentage of pre-disability income (replacement ratio), with limits to prevent over-insurance.
- Definitions of disability vary: own occupation, suited occupation, any occupation, or based on Activities of Daily Living (ADLs).
- Benefit features include waiting/deferred periods, escalation (before/in-claim), waiver of premium, rehabilitation/proportionate benefits, and expiry age.
- No surrender or maturity value; partial benefits possible for part-time return to work.
- Group IP is used by employers for statutory/discretionary sick pay, with options for continuation on leaving employment.

Critical Illness (CI) Cover

- Pays a lump sum on diagnosis of a specified illness or event; not indemnity-based.
- Product structures: stand-alone, rider, or accelerated benefit (pays on earlier of CI or death).
- Tiered benefits pay different amounts based on severity, allowing multiple claims and more nuanced cover.
- Benefits may include terminal illness, children’s cover, and TPD riders.
- Criteria for insurability: seriousness, frequency, clear definition, and sufficient data for pricing.
- Group CI is valuable where IP is unavailable, especially for blue-collar or high-risk occupations.

Long-Term Care Insurance (LTCI)

- Provides protection against the cost of home or nursing care, usually for the elderly or permanently disabled.

- Benefits can be indemnity (actual care costs) or cash (fixed income/lump sum), triggered by failure of ADLs or similar criteria.
- Product types: pre-funded (bought while healthy), immediate needs (for those already needing care).
- Structures: stand-alone, riders to life/CI/IP, unit-linked, tiered/severity-based payouts, and pension-linked solutions.
- Non-financial benefits may include assistive devices, alarms, and care advice.

Pricing and Assumptions

- Key assumptions: morbidity (inception, recovery, termination rates), mortality, expenses (initial, renewal, claims assessment), and investment returns.
- Margins for prudence are required due to high uncertainty in morbidity and expense experience.
- Tiered and severity-based products require multi-state modelling and careful definition of transition intensities.

Risks to the Insurer

- Morbidity risk: uncertainty in claim rates, parameter risk, anti-selection, moral hazard, and longevity risk (especially for IP and LTCI).
- Investment risk: mismatch between assets and liabilities, especially for long-term or unit-linked products.
- Expense risk: actual expenses may exceed loadings, especially for complex or low-volume products.
- Withdrawal/persistency risk: selective lapses can worsen the risk pool.
- Data risk: inaccurate or incomplete data can lead to mispricing and reserving errors.
- Operational, reputational, and regulatory risks, especially for LTCI and CI with complex definitions or political sensitivity.

Product Variations and Features

- Guaranteed vs. reviewable premiums: guarantees increase capital strain; reviewable rates can lead to selective withdrawals.
- No-claims discounts and GIOs (Guaranteed Insurability Options) are less common but may be offered.
- Unit-linked and with-profits structures shift some risk to policyholders and can improve capital efficiency.
- Benefit escalation, waiver of premium, and rehabilitation features are common in IP and LTCI.

Capital Requirements

- Driven by contract design (guarantees, unit-linking), initial expenses, premium frequency, and the gap between pricing and reserving bases.
- Solvency capital requirements are higher for health and care products due to morbidity uncertainty and long claim durations.
- LTCI capital needs are similar to endowments/whole life due to long-term premium payment and uncertain benefit timing.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Comparison of product structures** — Advantages/disadvantages of unit-linked vs. conventional IP, CI, or LTCI (e.g., capital efficiency, benefit flexibility, risk-sharing).
- **Product design and pricing for tiered/complex benefits** — How tiered CI or LTCI products differ in design, pricing, and risk compared to standard lump sum or indemnity products.
- **Impact of external events** — Effects of pandemics or regulatory changes on claims experience, pricing, and reserving (e.g., CI claims during/after lockdowns).
- **Customer needs analysis** — Identifying and explaining how product features meet specific customer needs (e.g., income replacement, debt repayment, asset protection).
- **Risk identification and management** — Detailed breakdown of morbidity, investment, expense, and withdrawal risks for new or modified products, and practical risk mitigation strategies (e.g., reinsurance, reviewable charges, underwriting).
- **Capital requirement drivers** — Explaining how product design, guarantees, premium frequency, and reserving bases affect capital needs for health and care products.
- **Reinsurance strategies** — Types of reinsurance suitable for claims risk in IP and CI, and how strategies differ for tiered vs. standard products.
- **Operational and legal considerations** — Importance of clear policy wording, claims definitions, and administrative processes to avoid disputes and regulatory issues.

Common Pitfalls & Exam Technique

- Omitting discussion of both advantages and disadvantages when comparing product structures (e.g., unit-linked vs. conventional).
- Failing to address all relevant risks (morbidity, investment, expense, withdrawal, data) for a given product scenario.
- Not linking product features directly to customer needs in answers.
- Overlooking the impact of guarantees and reviewability on capital requirements and risk.
- Neglecting to mention the complexity and data challenges of tiered/severity-based products.
- Ignoring the practicalities of claims management, underwriting, and policy administration in product design questions.
- Forgetting to discuss anti-selection, moral hazard, and selective withdrawal risks, especially for reviewable or flexible products.
- Not specifying the type of reinsurance or how it addresses the specific risk in question.
- Missing the need to discuss both pricing and reserving bases when explaining capital requirements.
- In scenario questions, always consider the impact of external events (e.g., pandemics, regulatory changes) on claims and operations.

#2 Assumptions HIGH PRIORITY

34 exam questions | last tested 2024 | recency 11.56

What You Must Know

This topic covers the principles, methodologies, and practical considerations for setting actuarial assumptions in life insurance, including their role in pricing, reserving, and embedded value calculations. Assumptions are a major source of risk and uncertainty, and their selection is central to actuarial work, making this a frequent and high-weighted exam topic.

Core Concepts

Assumptions as a Source of Risk

- Assumptions are made about uncertain future events (e.g., mortality, expenses, investment returns).
- Parameter risk arises because actual experience will differ from assumptions, potentially leading to financial loss.
- Assumptions cannot always be perfectly matched by assets, exposing insurers to risk.

Five-Step Framework for Setting Assumptions

- 1. Determine best estimate: Analyse historical data to set initial parameters.
- 2. Adjust for future expectations: Modify for anticipated changes in environment or business.
- 3. Balance credibility and judgment: Weigh data against professional judgment.
- 4. Incorporate margins for prudence: Add explicit margins based on risk and purpose.
- 5. Review and document: Ensure assumptions are justified and documented.

Mortality Assumptions

- Base mortality: Reflects expected future experience, considering target market, distribution, and underwriting.
- Data sources: Use internal experience, industry tables, reinsurer data, or population statistics as appropriate.
- Homogeneous grouping: Segment data by age, sex, smoker status, etc., ensuring credibility.
- Mortality trends: Allow for improvements or deteriorations using expectation, extrapolation, or explanatory approaches.
- Advanced modelling: Cohort effects, stochastic models (e.g., Lee-Carter), and predictive analytics may be used.

Morbidity Assumptions

- Income Protection (IP): Model transitions between healthy, sick, claiming, recovered, and dead states; estimate transition intensities by age, duration, and cause.
- Critical Illness: Estimate frequency for multiple conditions; sensitive to medical advances and data limitations.
- Long-Term Care Insurance (LTCI): Model both claim frequency and claim amount distributions, allowing for economic and medical factors.

Investment Return Assumptions

- Best estimate based on current market yields (e.g., redemption yield on matching assets).
- Adjust for future reinvestment risk, asset mix, taxation, and economic outlook.
- Margins for prudence are deducted to protect against lower-than-expected yields.
- Approach varies by valuation method: traditional (risk-free plus margin) vs. market-consistent (risk-free only).

Expense and Commission Assumptions

- Include both fixed and marginal costs; allocate per-policy expenses via policy fees or proportional loadings.
- Allow for expense inflation, typically linked to wage inflation and consistent with investment return assumptions.
- Use internal, industry, and public data (e.g., CPI, QES) to inform assumptions.

Withdrawal (Persistency) Assumptions

- Base on company's own experience or industry data; adjust for changes in market, distribution, or product features.
- Balance data credibility with professional judgment; include margins for uncertainty.

Allowing for Risk in Assumptions

- Risk discount rate (RDR): Add risk premium to discount rate to allow for uncertainty in profit emergence.
- Stochastic approach: Use probability distributions and simulations (e.g., Monte Carlo) to quantify risk and set margins.
- Explicit margins: Adjust individual assumptions directly for prudence.

Valuation Bases and Consistency

- Valuation basis reflects expected experience, margins, regulation, and consistency requirements.
- Supervisory reserves aim for policyholder security; published accounts may use best estimate or prudent bases.
- Internal management accounts use realistic (best estimate) assumptions for profitability and performance management.
- Consistency is required between assumptions for assets and liabilities, and between pricing, reserving, and embedded value calculations.

Best Estimate and Embedded Value (EV)

- Best estimate reserves: Use realistic assumptions without margins; negative reserves allowed.
- Embedded value: Sum of shareholder net assets and present value of future shareholder profits (PVFP), discounted at a risk discount rate or with explicit risk margin.
- EV basis must be consistent with reserving basis; risk is allowed for via RDR or market-consistent approaches.

Pricing vs Reserving Assumptions

- Pricing assumptions are more uncertain (new business, untested markets); reserving assumptions benefit from known portfolio and experience.
- Statutory reserving basis is generally more prudent than pricing basis (higher mortality, lower investment return, higher expenses).
- Using pricing assumptions for reserving is only appropriate if margins are sufficient (e.g., with-profits business).

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Describing and justifying the process for setting key assumptions** — Candidates must outline step-by-step how to set assumptions for mortality, morbidity, expenses, investment return, and persistency, referencing data sources, credibility, and adjustments for future expectations.
- **Application to product-specific scenarios** — Questions often require setting or reviewing assumptions for new or changed products (e.g., cancer-only CI, direct whole life, unit-linked endowment), including challenges and data limitations.
- **Embedded value (EV) calculations** — Candidates must describe EV components, explain the principles for setting the EV basis, and outline how to determine assumptions for each key item (e.g., future profits, net assets, risk margins).
- **Impact of regulatory or market changes** — Examiners test understanding of how changes (e.g., bans on medical underwriting) affect assumptions, risk, and experience over time.
- **Risk mitigation and prudence** — Candidates are expected to discuss methods for allowing for risk (RDR, stochastic, explicit margins) and the rationale for prudence in pricing vs reserving.
- **Consistency and reconciliation between bases** — Questions probe the need for consistency between pricing, reserving, and EV bases, and the implications of mismatches.
- **Listing and explaining key parameters** — Candidates must identify all relevant assumptions for a given product and explain their significance and how they would be set.

Common Pitfalls & Exam Technique

- Omitting key parameters (e.g., only listing mortality and ignoring expenses, lapses, investment return, etc.).

- Failing to distinguish between best estimate and prudent (statutory) assumptions, or between pricing and reserving bases.
- Not addressing both data-driven and judgmental aspects of assumption setting, or neglecting to mention margins for prudence.
- Ignoring the need for consistency between assumptions (e.g., inflation and investment return, or between asset and liability bases).
- Overlooking the impact of external changes (regulation, market, medical advances) on assumptions and experience.
- Not explaining the rationale for risk margins or the methods for quantifying risk (RDR, stochastic, explicit margins).
- In EV questions, failing to describe both the net asset and PVFP components, or not explaining how risk is allowed for.
- For product-specific questions, not tailoring assumptions to the product's features, distribution, or target market.
- Neglecting to mention data limitations and the need for professional judgment where data is sparse or unreliable.
- Not structuring answers logically (e.g., not following the five-step framework or not separating base rates from trends).

#3 Risks HIGH PRIORITY

32 exam questions | last tested 2025 | recency 13.63

What You Must Know

This topic covers the full spectrum of risks faced by life insurance companies — their sources, interactions, and practical management. It spans risk identification (model, parameter, and operational risks across mortality, investment, expense, and withdrawal dimensions) through to hands-on management techniques including expense control, persistency management, new business monitoring, and options/guarantee management. This is a high-frequency exam area because it underpins all actuarial work in pricing, reserving, and capital management.

Core Concepts

Sources of Risk for Life Insurers

- Policy and other data: Inaccurate or incomplete data can lead to mispricing, inadequate reserving, and poor decision-making.
- Mortality and morbidity: Uncertainty in death, illness, or disability rates affects claims and reserves.
- Investment performance: Variability in asset returns, mismatching, and market volatility impact the ability to meet guarantees and liabilities.
- Expenses and inflation: Actual expenses may exceed assumptions due to inflation, inefficiency, or business volume changes.
- Withdrawals (lapses, surrenders): Policyholder behaviour can deviate from assumptions, affecting income, expenses, and risk exposure.
- Mix of new business: Changes in product type, size, or distribution channel alter risk and profitability profiles.

- Volume of business: Too high or too low volumes create capital strain or threaten expense recovery and viability.
- Guarantees and options: Embedded product features introduce additional risk and capital strain.
- Strategic, operational, legal, regulatory, and counterparty risks: External and internal factors can disrupt operations or financial stability.

Data Risks

- Quality, completeness, and accuracy of policy data are critical for reliable actuarial work.
- Model point representation and model risk: Simplifications may misrepresent the business.
- Credibility, relevance, and quality of data used for assumption setting are often limited, especially for new or niche products.
- Health and care products face severe data limitations due to low volumes and heterogeneity.

Mortality and Morbidity Risks

- Model risk: The chosen statistical model may not reflect reality (e.g., assuming i.i.d. lives).
- Parameter risk: Estimated rates may not match actual experience.
- Random fluctuations: Especially significant in small portfolios.
- Direction of risk: Higher mortality is a risk for life cover; longevity is a risk for annuities.
- Selection risks: Anti-selection and selective withdrawals can worsen experience.
- Health and care products: Risks focus on claim incidence and recovery rates, compounded by data limitations.

Investment Risks

- Model, parameter, and random fluctuation risks apply to investment returns.
- Mismatching risk: Asset-liability duration mismatch leads to reinvestment or capital value risks.
- Investment guarantees: Risk that assets underperform relative to guaranteed benefits.

Expense, Inflation & Cost Control

- Expense risk arises from actual expenses exceeding assumptions due to inflation, inefficiency, or business volume changes; inflation risk can erode margins, especially for level premium products.
- Expense risk is linked to new business mix, withdrawal rates, and investment performance (for unit-linked charges).
- Expense control aims to keep actual expenses and commissions within premium loadings; insurers monitor by category (initial, renewal, investment) and benchmark against competitors.

- Methods to control expenses: increasing volumes to spread fixed costs, automation and process improvements, cheaper distribution channels, and managing staff costs.
- Commissions are controlled at product line, distribution channel, and individual salesperson/broker level — increases often start at the individual level and must be caught early.
- To manage mismatches between expense incidence and loadings, insurers may reprice products, alter product mix, change distribution channels, or adjust minimum premiums.

Withdrawal, Persistency & Retention

- Model, parameter, and random fluctuation risks in predicting lapses, surrenders, and paid-up rates.
- Withdrawals affect mortality and expense assumptions through selective behaviour and fixed cost spreading; incorrect assumptions lead to immediate losses from unrecovered acquisition costs.
- Persistency rates are monitored by distribution channel and salesperson/broker, with pricing and valuation assumptions updated as experience evolves.
- Improving persistency: aligning sales incentives (lower initial, higher renewal commissions, clawbacks), financial underwriting, and addressing root causes (customer service, payment methods, product design, staff training).
- Product features that help reduce lapses: premium holidays, capping premium increases, fair alteration terms, and surrender penalties.

Mix, Volume & New Business Strain

- Mix by nature, size, and source affects capital requirements, expense recovery, and risk profile.
- Too high volumes: Capital strain, operational overload, and potential solvency issues. Too low volumes: Failure to recover fixed costs, loss of competitiveness, and long-term viability threats.
- New business strain arises from acquisition expenses, capital requirements, and mismatches between income and outgo; managed through less onerous product design (unit-linked, reviewable premiums, reduced guarantees).
- Monitoring is done by product line and distribution channel, tracking contract numbers and premium amounts.
- Insurers must ensure sufficient capital and administrative resources to support planned growth in new business volumes.

Options, Guarantees & Their Management

- Mortality/morbidity and investment guarantees introduce anti-selection, capital strain, and model/parameter risks.
- Options (surrender, premium holidays, guaranteed insurability) increase uncertainty and potential for adverse selection.
- Insurers monitor the cost of options and guarantees against loadings, adjusting charges or product design if losses emerge.

- Benefits of options (marketability, customer need) must be weighed against costs (actual/potential losses, risk exposure).
- Mitigation: prudent reserving, hedging with financial derivatives, buying back options from policyholders, or restructuring product design.
- Changes to options/guarantees are easier for new business; for in-force policies, strict policy wording and fair policyholder treatment are required.

Strategic, Operational & Counterparty Risks

- Strategic risk: Board decisions may conflict with actuarial advice or policyholder interests.
- Operational risk: Failures in management systems, controls, or fraud can cause financial loss.
- Counterparty risk: Reinsurer, outsourcer, or investment default exposes the insurer to loss.
- Legal, regulatory, and fiscal risks: Changes or non-compliance can have severe financial and reputational impacts.

Aggregation & Concentration of Risk

- Risks must be considered in aggregate, allowing for correlations and interactions (e.g., via stochastic modelling).
- Concentration risk arises from large exposures or non-independent risks (e.g., catastrophe, asset over-concentration).
- Credit rating downgrades can restrict business and increase lapse risk.

Risks from Distributors

- Distributors can drive lapses, exploit product features, create credit risk, or cause mis-selling and legal issues.

Risk Management Framework & Actuary's Role

- Risk must be measured relative to company aims, capital, and regulatory requirements; strategic decisions balance risk and return.
- Actuaries regularly advise directors on the nature, size, and management of risks, including scenario modelling for sensitive risks.
- They document, implement, and monitor risk management strategies, supporting an enterprise risk management (ERM) approach that captures aggregate risk and diversification benefits.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Product-specific risk analysis** — Identifying and explaining the main risks (mortality, morbidity, investment, expense, withdrawal) for a given product design, including new features or changes.
- **Impact of product changes** — Assessing how new guarantees, options, or changes in charging structure increase risk and how these can be managed.
- **Expense and persistency risk linkage** — Explaining how lapses/withdrawals affect expense recovery and overall profitability.

- **Expense and commission management** — Discussing how to monitor and control expenses and commissions at multiple organisational levels, and the impact of distribution channel choice.
- **Managing persistency and policyholder behaviour** — Suggesting ways to improve persistency and reduce lapses, including incentive structures and product design features.
- **Investment risk and mismatching** — Discussing risks from asset-liability mismatches, especially with guarantees or overseas investments.
- **Options and guarantees risk management** — Identifying risks from product features and outlining methods to monitor, price, and mitigate these risks.
- **Managing new business strain and capital requirements** — Explaining how product design and business mix affect capital strain and how to manage it.
- **Risk mitigation strategies** — Proposing practical measures: product design changes, underwriting, reinsurance, reserving, and monitoring actual vs. expected experience.
- **Operational and counterparty risks** — Recognising risks from outsourcing, administration, or distributor actions.
- **Strategic and regulatory risks** — Identifying risks from regulatory changes, compliance, or strategic decisions.

Common Pitfalls & Exam Technique

- Omitting less obvious risks (data, operational, counterparty, regulatory) when asked for a comprehensive risk analysis.
- Failing to link risks to specific product features or changes (e.g., how a new guarantee increases investment risk).
- Not distinguishing between model, parameter, and random fluctuation risks for mortality, investment, or withdrawal assumptions.
- Overlooking the dynamic links between risks (e.g., how lapses affect expense and mortality risk).
- Omitting discussion of both expense and commission control, or failing to address monitoring at multiple organisational levels.
- Not linking risk mitigation strategies to the specific product or risk in question — generic answers score poorly.
- Ignoring the challenges in setting assumptions for new products, such as lack of credible data or differences from existing products.
- Forgetting to mention the impact of distribution channel on expenses, persistency, and anti-selection risk.
- Neglecting to discuss both the benefits and costs of options/guarantees, or the time lag in risk mitigation for in-force business.
- Not addressing both the direction and magnitude of risk (e.g., higher mortality vs. longevity risk for different products).
- Failing to consider capital and administrative resource constraints when discussing new business growth.
- Not referencing the actuary's ongoing role in risk identification, quantification, and communication to management.
- In "outline" questions, writing too much detail or missing breadth — use concise, well-structured bullet points covering all relevant angles.

#4 Life insurance contracts HIGH PRIORITY

29 exam questions | last tested 2025 | recency 14.96

What You Must Know

This topic covers the structure, features, customer needs, and insurer risks of the main types of life insurance contracts, including endowment, whole life, term assurance, annuities, unit-linked, and index-linked products. It is a core exam area because it underpins product design, risk management, and capital requirements in life insurance practice.

Core Concepts

Life Insurance Contract Types & Definitions

- Life insurance contracts pay benefits contingent on human life (death or survival) in exchange for premiums.
- Main contract types: endowment assurances, whole life assurances, term assurances (including convertible/renewable), annuities (immediate and deferred), unit-linked, and index-linked contracts.

Endowment Assurances

- Pay a benefit on survival to a known date (maturity) or on earlier death.
- Meet savings (retirement, loan repayment) and protection needs.
- Surrender values usually available and increase over time; not necessarily linked to death/maturity benefit.
- Structures: without-profits (guaranteed), with-profits (bonuses), unit-linked (investment performance).
- Risks: investment (varies by structure), mortality (especially if death benefit is high), expense (inflation, low sales), withdrawal (negative asset share, selective lapses), group risks (concentration).
- High capital requirements due to large reserves, initial expenses, and prudent reserving.

Whole Life Assurances

- Pay a benefit on death whenever it occurs; no fixed term.
- Primarily protection (income replacement, funeral, tax planning), but can serve as savings (with-profits/unit-linked).
- Surrender values rare in some markets; paid-up options possible.
- Risks: mortality (high for older/new policies), anti-selection, investment (especially non-linked), expense/inflation, withdrawal (negative asset share).
- High capital requirements due to long-term reserves and initial expenses.

Term Assurances

- Pay a benefit on death within a specified term; no payment if survival to term end.
- Meet protection needs (dependants, loan cover, keyman, group schemes).

- Generally no surrender value; low reserves.
- Risks: mortality (claims > expected), anti-selection, expense (initial/fixed costs, inflation), withdrawal (especially for decreasing term), random fluctuation (epidemics).
- Low product design capital, but high initial expense and solvency margin requirements.

Convertible and Renewable Term Assurances

- Allow conversion to permanent cover or renewal without further medical evidence.
- Meet needs for affordable, flexible protection with future certainty.
- No surrender value before conversion/renewal.
- Risks: heightened anti-selection at option exercise, renewal risk > conversion risk, option cost management (extra premium, possible limited testing).
- Higher capital/reserving needs due to option guarantees and uncertainty at exercise.

Annuity Contracts

- Pay regular benefits on survival (immediate: payments start now; deferred: payments start at vesting date).
- Structures: conventional without-profits (guaranteed), index-linked (inflation protection), with-profits (bonuses), unit-linked (investment performance).
- Immediate annuities: single premium, no surrender value, meet needs for lifetime income, longevity protection, and dependants' income.
- Deferred annuities: regular/single premiums, may allow surrender during deferment, meet retirement savings and longevity protection needs.
- Risks: longevity (annuitants live longer), anti-selection, investment (matching assets to uncertain liabilities), expense (inflation, operational), withdrawal (deferred phase), group risks.
- Capital requirements: as for endowments (deferred phase), plus reserves for annuity guarantees and options.

Unit-Linked Contracts

- Benefits linked to investment fund performance; policyholder bears most investment risk.
- Premiums buy units; charges deducted via allocation rates, fund charges, or unit cancellations.
- Benefits: bid value of units (may have minimum guarantees); surrender value = bid value minus penalty.
- Unit fund (policyholder's value) vs. non-unit fund (insurer's charges/expenses/profit).
- Meet needs for flexibility (investment choice, premium structure, cover), inflation protection, savings, wealth transfer, group benefits.
- Risks: expense (charges may not match costs), investment (poor returns, guarantee shortfalls), mortality (guaranteed death benefits), withdrawal (negative asset share, selective lapses), external (reputation, regulation).
- Capital requirements: driven by guarantees, initial expenses, and reserving for negative non-unit reserves.

Index-Linked Contracts

- Benefits linked to an economic index (e.g., CPI) to provide inflation protection.
- Meet needs for real benefit security and standard of living maintenance.
- Risks: investment (matching assets to index, mismatching, strategy risk), practical asset availability.

Risks to Policyholders

- Vary by product: investment/market risk, inflation risk, counterparty (insurer insolvency) risk, flexibility/need risk.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Matching product features to customer needs** — Clearly link contract structures (e.g., deferred annuity, unit-linked endowment) to specific financial needs (retirement income, inflation protection, flexibility, wealth transfer).
- **Detailed risk analysis for insurers** — Identify and explain all key risks (mortality, longevity, investment, expense, withdrawal, anti-selection, option risk) for each contract type, including how risks differ by product structure and how they can be managed (e.g., underwriting, reinsurance, reserving, asset-liability matching).
- **Capital and reserving requirements** — Discuss how product design, initial expenses, guarantees, and reserving/pricing basis differences drive capital needs for each contract.
- **Asset-liability matching** — For annuities and index-linked products, describe liability characteristics and recommend suitable backing assets, considering practical constraints.
- **Product design changes and their impact** — Evaluate how new features (e.g., enhanced guarantees, flexible withdrawals, new charging structures) increase insurer risks and suggest risk management strategies.
- **Comparison of product types** — Contrast unit-linked vs. conventional products from both policyholder and insurer perspectives (risk, flexibility, charges, guarantees).
- **Charges and expense recovery** — List and explain different charging methods and how initial expenses can be recouped under various product designs.
- **Option-pricing for guarantees** — Explain how market option-pricing techniques can be used to value embedded options (e.g., guaranteed annuity options).

Common Pitfalls & Exam Technique

- Omitting key risks (e.g., anti-selection, expense inflation, selective withdrawal) or failing to tailor risk explanations to the specific contract structure.
- Not linking product features directly to customer needs in answers—always state how the contract meets specific financial objectives.

- Confusing unit fund and non-unit fund concepts in unit-linked products; be precise about which risks/charges apply to each.
- Neglecting to discuss capital requirements and how they are affected by product design, guarantees, and reserving basis.
- For asset-liability matching, failing to consider practical constraints (e.g., asset availability, reinvestment risk) and the impact of mismatching.
- Overlooking the impact of regulatory, reputational, and market risks, especially for unit-linked and index-linked products.
- When discussing new product features, not fully exploring all new risks introduced and how they can be managed (e.g., via reinsurance, reserving, option pricing, or product design changes).
- Not providing enough detail on how charges are structured and how initial expenses are recovered, especially for unit-linked and with-profits products.
- Failing to address both insurer and policyholder perspectives where required.

#5 Asset shares and profit distribution MEDIUM PRIORITY

27 exam questions | last tested 2025 | recency 11.44

What You Must Know

This topic covers the calculation and use of asset shares in life insurance, and the various methods for distributing profits (bonuses) to with-profits policyholders. It is central to understanding how insurers manage surplus, set bonus rates, and balance policyholder and shareholder interests—making it a frequent focus in the exam.

Core Concepts

Asset Share: Definition and Components

- Asset share is the accumulated value of all premiums received, rolled up with actual investment returns, less deductions for expenses, claims, tax, and other outgoes.
- It is similar to a gross premium retrospective reserve and is calculated per policy in force.
- Monies In: Premiums, investment returns (including unrealized gains), and allocated profits from other business lines.
- Monies Out: Commissions, expenses (initial and renewal), cost of claims (death strain at risk), tax, shareholder transfers, cost of capital, and contributions to free assets.

Asset Share Calculation: Recursive Formula

- Calculated recursively year by year, starting from zero at inception.
- Formula:

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AS_t+1 = \frac(AS_t + P - E)(1 + i) - q'_x+t · S1 - q'_x+t
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where AS = asset share, P = premium, E = expenses, i = investment return, S = sum assured, q' = actual mortality rate.

- Claims are assumed paid at year-end; only the death strain at risk (benefit in excess of asset share) is deducted, not the full sum assured.
- Asset share development phases: initial negative (new business strain), growth (compounding returns), and product-specific trajectories (e.g., endowment vs. term assurance).

Profit Distribution Methods

- Additions to Benefits (Conventional With-Profits): Profits distributed via regular reversionary bonuses (simple, compound, super-compound), special reversionary bonuses, and terminal bonuses.
- Accumulating With-Profits: Bonuses added to a fund value, often unitised; regular bonuses act like interest, with terminal bonuses aligning payout to asset share.
- Revalorisation Method: Benefits increased annually by a proportion of actual investment income above a guaranteed technical rate; highly formulaic, transparent, and offers little smoothing or discretion.
- Contribution Method: Surplus distributed according to each policy's contribution to investment, mortality, and expense profits; policyholders share in all sources of surplus and risk.

Bonus Types and Features

- Regular reversionary bonuses: Declared annually, become guaranteed, and may be simple, compound, or super-compound (the latter defers surplus and is more capital efficient).
- Special reversionary bonuses: One-off, used for non-recurring surplus sources.
- Terminal bonuses: Paid only at claim (maturity, death, surrender), not guaranteed until event, used to distribute volatile or residual surplus and align payout to asset share.
- Market Value Reduction (MVR): Discretionary reduction on surrender to ensure payout does not exceed asset share, protecting remaining policyholders.

Smoothing, Discretion, and Policyholder Expectations

- Smoothing: Spreads profits over time to reduce volatility in payouts; more feasible under additions to benefits than revalorisation.
- Discretion: Actuarial discretion in bonus setting allows for management of PRE (Policyholders' Reasonable Expectations) and financial stability.
- PRE: Developed from company practice, marketing, and market norms; failure to meet PRE can lead to complaints and regulatory action.

Business and Capital Considerations

- Delaying profit distribution increases free assets, improves solvency, and allows greater investment flexibility.
- Choice of bonus method affects guarantees, smoothing, and capital requirements.

- In proprietary companies, a portion of surplus (often 1/9th) is transferred to shareholders, creating a balance between policyholder and shareholder interests.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Features and mechanics of profit distribution methods** — Clearly outline the key features, advantages, and disadvantages of methods like additions to benefits, revalorisation, and contribution method. Be able to compare them directly.
- **Calculation of asset shares and dividends** — Apply the recursive formula or contribution method to calculate asset shares or policyholder dividends, using actual and expected experience for investment, mortality, and expenses.
- **Types and roles of bonuses** — Describe and distinguish between regular reversionary, special reversionary, and terminal bonuses, including their impact on guarantees and capital.
- **Rationale for bonus philosophy and split** — Discuss factors influencing the split between reversionary and terminal bonuses, especially in the context of asset volatility, capital strength, and product design.
- **Impact of asset mix and investment strategy** — Relate the choice of assets backing with-profits liabilities to the nature of guarantees, smoothing, and bonus setting, and comment on the suitability of different asset mixes.
- **Reasons for deferring or accelerating profit distribution** — Explain why insurers might delay surplus distribution (e.g., to improve solvency, manage volatility, or support new business strain).
- **Policyholder expectations and communication** — Discuss how PRE is formed and managed, and the risks of failing to meet expectations.
- **Business objectives and conflicts** — Analyze the tension between maximizing policyholder bonuses, maintaining solvency, and meeting shareholder interests.

Common Pitfalls & Exam Technique

- Omitting key deductions or income items in asset share calculations (e.g., tax, cost of capital, shareholder transfers).
- Confusing the features of different surplus distribution methods—be precise about which method is being described or calculated.
- Failing to link bonus types to their impact on guarantees, smoothing, and capital (e.g., why terminal bonuses are used for volatile returns).
- Not addressing both policyholder and shareholder perspectives in proprietary company scenarios.
- Neglecting to discuss PRE and the importance of communication in bonus setting questions.
- Overlooking the impact of asset mix on the ability to meet guarantees and smooth bonuses.
- In calculation questions, not showing all steps or not clearly separating investment, mortality, and expense surplus components.

- For scenario/discussion questions, not considering both short-term and long-term implications (e.g., for closed books or after market shocks).

#6 Unit pricing and actuarial funding HIGH PRIORITY

26 exam questions | last tested 2025 | recency 13.61

What You Must Know

This topic covers the principles, mechanics, and risks of unit pricing and actuarial funding in unit-linked life insurance products. It is frequently examined because it underpins the fair treatment of policyholders, the management of insurer capital strain, and the practical operation of modern investment-linked contracts.

Core Concepts

Unit Pricing

- Unit-linked benefits depend on the value of units, which reflect the performance of underlying assets.
- Unit pricing is the process of determining the value of assets attributable to each unit in the fund.
- Bid and offer prices apply to both assets and units, and also refer to the basis used for pricing units.

Principal Risks in Unit Pricing

- Timing risk: Unit prices are set at discrete intervals, but asset values change continuously.
- Asset value volatility and sudden market movements can cause pricing mismatches.
- Outdated or approximate asset valuations (especially for illiquid assets like property).
- Tax on unrealised gains, accrued interest, and calculation errors.
- Incorrect pricing basis (appropriation vs expropriation), regulatory changes, and reputational risk from inequitable pricing.
- Anti-selection if unit prices overstate asset values.

Appropriation and Expropriation Prices

- Appropriation price: Price at which units are created, based on the offer price of assets plus purchase expenses and taxes.
- Expropriation price: Price at which units are cancelled, based on the bid price of assets minus sale expenses.
- These prices ensure equity between entering, exiting, and continuing policyholders.

Fund Pricing Bases

- Offer basis: Used for expanding funds (net creation of units).
- Bid basis: Used for contracting funds (net cancellation of units).

- Switching between bases is only done for significant, sustained cashflow changes to maintain equity.

Pricing for Policyholders

- Offer price: Price at which policyholders buy units from the insurer.
- Bid price: Price at which insurer buys units back from policyholders.
- The bid-offer spread covers initial charges, management expenses, and profit.
- Rounding of prices can favour either the insurer or the customer, subject to regulation.

Management Box

- Insurers may hold a management box (excess units) to manage daily transactions without frequent asset trades.
- This reduces volatility but exposes the insurer to investment risk on the box holdings.

Actuarial Funding

- Technique to reduce initial capital strain by holding a reserve lower than the full unit value, taking credit for future management charges.
- Reserve = Normal reserve minus EPV of future initial expense charges recouped via management charges.
- Commonly used when initial expenses are high and recouped over time via fund management charges.

Mechanism of Actuarial Funding

- Instead of fully funding the unit reserve, the insurer holds the actuarial present value of the unit fund, discounted at a rate less than the fund management charge.
- Funding factor (e.g., $A_{x+t:\overline{n-t|}}$) is applied to the unit fund to determine the reserve.
- The insurer "borrows" units and repays them using future management charges, eliminating investment risk in the buy-back process.

Conditions for Use of Actuarial Funding

- High and regular management charges must be present to fund the buy-back of units.
- Surrender penalties must ensure the reserve is never less than the surrender value.
- Future net cashflows must remain positive after management charges are reduced to rebuild the reserve.
- Benefit must be contingent (e.g., payable on death or maturity).

Absence of Investment Risk in Actuarial Funding

- Both the liability (missing units) and the funding source (management charges) are linked to the unit price, so investment performance does not affect the ability to buy back units.

- The insurer is exposed to lapse and expense risk, but not investment risk in this context.

Effect on Cashflows and Reserves

- At inception, actuarial funding creates a positive cashflow to the non-unit fund (difference between premium and funded reserve).
- Subsequent years see reduced net cashflows as management charges are used to buy back units.
- Reserves under actuarial funding start lower and converge to the standard reserve as the policy matures.
- On death, the non-unit fund must cover the difference between units held and the full benefit.

Basic Equity Principle

- Unit pricing should ensure that transactions by one policyholder do not affect the value of units held by others.
- Prices should only reflect asset performance and allowable charges, not the creation or cancellation of other units.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Definitions and Principles** — Define appropriation/expropriation prices, actuarial funding, and the basic equity principle, and explain their rationale.
- **Calculation of Unit Prices** — Calculate appropriation, expropriation, and offer prices using provided fund data, including treatment of expenses, taxes, and rounding.
- **Application of Equity Principle** — Explain how pricing methods preserve equity between policyholders and prevent cross-subsidies.
- **Risks in Unit Pricing and Product Changes** — Identify and discuss risks arising from product features, pricing bases, and operational changes (e.g., new guarantees, withdrawal options, outsourcing).
- **Conditions and Limitations of Actuarial Funding** — List and explain the conditions required for prudent actuarial funding, including surrender penalties and management charge sufficiency.
- **Impact of Surrender Penalties** — Discuss how surrender penalty design affects the insurer's ability to actuarially fund and the fairness to policyholders.
- **Effect on Cashflows and Reserves** — Describe and illustrate how actuarial funding alters cashflows, reserve emergence, and profit recognition over time.
- **Equity Across Policyholder Generations** — Provide examples of how unit pricing can affect fairness between different cohorts of policyholders.
- **Practical Application** — Apply principles to real-world scenarios, such as expanding/contracting funds, and justify the choice of pricing basis.

Common Pitfalls & Exam Technique

- Confusing the calculation basis for appropriation (offer) and expropriation (bid) prices — always specify which basis and include all relevant expenses and taxes.
- Omitting the rationale for the basic equity principle or failing to link it to the prevention of cross-subsidies.
- Neglecting to discuss the impact of surrender penalties on actuarial funding and policyholder fairness.
- Forgetting to state assumptions in calculations (e.g., treatment of cash, rounding, or timing of charges).
- Overlooking the need for positive future cashflows and sufficient management charges when discussing actuarial funding.
- Not distinguishing between investment risk (eliminated in actuarial funding) and other risks (lapse, expense, operational).
- Failing to explain how unit pricing decisions affect different generations of policyholders or the insurer's capital requirements.
- In calculation questions, not showing all steps clearly or not rounding as instructed.

#7 Monitoring experience MEDIUM PRIORITY

25 exam questions | last tested 2025 | recency 10.60

What You Must Know

This topic covers the reasons, methods, and implications of monitoring actual experience in a life insurance company. It is a frequent exam focus because it underpins the actuarial control cycle, informs assumption setting, and drives corrective actions in product design, pricing, operations, and risk management.

Core Concepts

Reasons for Monitoring Experience

- Provides management information for decision-making, profitability analysis, and performance measurement (e.g., remuneration, identifying profitable products/channels).
- Enables updating of pricing and reserving assumptions to reflect actual experience and emerging trends.
- Supports the actuarial control cycle by feeding back results to refine models and business strategies.
- Allows early identification of adverse trends, prompting corrective actions (e.g., re-pricing, product redesign, changes in underwriting or reinsurance).
- Assists in regulatory compliance and reporting requirements.

Corrective Actions Following Adverse Experience

- Re-pricing or redesigning products to restore profitability.
- Changing investment, sales, or reinsurance strategies.
- Adjusting commission structures, distribution channels, or underwriting processes.

- Modifying profit distribution (e.g., bonus rates) or operational resources (e.g., workforce, IT systems).

Management Information from Monitoring

- Identification of profitable/unprofitable products, channels, and markets.
- Assessment of operational efficiency and investment strategy success.
- Insight into business mix, sales trends, and customer behavior.

Data Requirements for Meaningful Monitoring

- Stable, consistent, and credible data volumes, grouped by homogeneous risk factors.
- Balance between granularity (homogeneity) and statistical credibility.
- Appropriate time intervals: e.g., 3–4 years for mortality, 12 months for expenses.
- Recent data preferred for current trends, but longer periods may be needed for credibility.

Grouping Factors in Experience Analysis

- Mortality: contract type, age, sex, duration, smoker status, underwriting, location, occupation.
- Persistency: contract type, duration, sales method, premium frequency/size, collection method, term, age, sex.
- Expenses: split by function (initial, renewal, termination, investment), product line, accounting fund.
- Investment: asset type, fund, product type.

Role of Big Data

- Enables refined risk classification and more accurate individual rating.
- Identifies preferred risks and can influence customer behavior (e.g., health advice).

Types of Experience Investigations

- Demographic: mortality, morbidity, persistency (withdrawal rates).
- Financial: expenses, investment returns.

Process for Mortality Investigation

- Collect and reconcile data; group into homogeneous cells.
- Choose appropriate time interval for credibility.
- Calculate crude rates, graduate to smooth, compare to standard tables.
- Perform actual vs expected (A/E) analysis; check reasonableness.
- Take corrective actions if adverse trends are found (e.g., reprice, update assumptions, adjust bonuses).

Process for Persistency (Withdrawal) Investigation

- Collect and group data by relevant risk factors.

- Calculate persistency rates (first year and subsequent years) using appropriate denominators.
- Regroup if necessary for credibility; analyze other discontinuance forms (paid-up, partial withdrawals).
- Interpret results in context of economic and competitive environment.

Process for Expense Investigation

- Use a 12-month period; exclude commission.
- Split expenses by function and analyze cost drivers (per policy, per premium, per benefit).
- Allocate overheads (salaries, property, IT) pragmatically.
- Amortize large or exceptional items; reconcile to accounts and previous investigations.

Process for Investment Return Investigation

- Choose valuation basis appropriate to purpose (e.g., book value for with-profits, market value for unit-linked).
- Subdivide by asset type, fund, and return component; analyze gross and net of expenses.
- Benchmark against internal assumptions, indices, and competitors.
- Feed results into control cycle (investment strategy, surplus distribution, pricing, reserving).

Analysis of Surplus

- Surplus is the change in free assets (assets less liabilities) over a period.
- Sources: differences between actual and assumed experience (investment, mortality, expenses, withdrawals).
- Valuation basis affects interpretation: prudent basis should yield positive surplus; best estimate may yield positives and negatives.

Using Experience Analysis Results

- Product & Pricing: update pricing, revise design, change mix, adjust underwriting/reinsurance, improve policy wording.
- Operational & Distribution: improve retention, marketing, sales, commission, staffing.
- Actuarial & Risk Management: improve models, update reserving, enhance governance, upgrade systems/data.
- Capital & Investment: adjust investment strategy, bonus distribution, capital raising/allocation.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Stating and explaining reasons for monitoring experience** — Candidates must list and elaborate on the main purposes, including management information, updating assumptions, and identifying adverse trends (2020 JUNE Q2, 2024 NOV Q3).

- **Describing the process for specific experience investigations** — Detailed steps for mortality, persistency, expense, and investment return analyses, including data collection, grouping, calculation, and interpretation (2020 JUNE Q2, 2019 NOV Q1, 2024 NOV Q3).
- **Identifying corrective actions following adverse experience** — Suggesting practical responses such as re-pricing, product redesign, operational changes, and updating assumptions (2019 NOV Q1, 2024 NOV Q3).
- **Discussing data requirements and grouping** — Emphasis on homogeneity, credibility, time intervals, and grouping factors for meaningful analysis (2020 JUNE Q2, 2019 NOV Q1).
- **Applying experience monitoring to product and operational scenarios** — Analyzing how monitoring informs product changes, risk management, and operational decisions (2025 NOV Q2, Q3, 2023 NOV Q6).
- **Explaining the impact of experience on surplus and embedded value** — Understanding sources of surplus and how experience analysis feeds into surplus/EV reporting (2024 NOV Q3, 2019 NOV Q1).
- **Linking monitoring to the actuarial control cycle** — Demonstrating how experience feedback leads to assumption and process updates (2019 JUNE Q1, 2020 JUNE Q2).

Common Pitfalls & Exam Technique

- Omitting key reasons for monitoring (e.g., only mentioning assumption updates, not management information or control cycle feedback).
- Failing to provide sufficient detail in process descriptions (e.g., not specifying grouping factors or time intervals).
- Confusing corrective actions with monitoring itself — always link actions to specific findings from experience analysis.
- Neglecting the importance of data quality, homogeneity, and credibility in investigations.
- Not tailoring answers to the product or scenario in the question (e.g., missing unique risks or operational implications).
- Overlooking the need to reconcile and validate results against accounts, previous investigations, and industry benchmarks.
- Forgetting to mention both demographic and financial investigations where relevant.
- In surplus analysis, not distinguishing between prudent and best estimate bases or not explaining the implications of negative results.
- When asked for actions, not covering the full spectrum: product, operational, actuarial, and capital/investment responses.

#8 Underwriting & data checks MEDIUM PRIORITY

25 exam questions | last tested 2023 | recency 8.39

What You Must Know

This topic covers the principles and practicalities of underwriting and data checks in life insurance, focusing on how insurers assess risk, set terms, and ensure data

integrity for pricing, reserving, and reporting. It is frequently examined because robust underwriting and data validation are critical to managing anti-selection, ensuring fairness, and maintaining financial soundness.

Core Concepts

Data Checks

- Reconciliation of key data items (e.g., sum assured, premiums, policy counts) between periods and with accounting records.
- Analysis of business movements: new business, lapses, claims, alterations, and their impact on in-force data.
- Calculation and review of ratios: lapse rates, benefit paid rates, new business rates, expense and claims ratios.
- Consistency checks: compare averages (e.g., sum assured, premiums) and ratios to prior periods and expectations.
- Spot checks and sensitivity tests on key parameters to ensure reasonableness and identify anomalies.
- Identification of unusual or impossible values (e.g., extreme sums assured, impossible dates).
- Reconciliation of claims data with revenue accounts and tie-backs to financial statements.
- Use of embedded value profit and surplus analysis to explain changes in results.

Underwriting: Definition and Purpose

- Underwriting is the process of assessing insurance risk to determine acceptability, premium, and terms.
- Key purposes: manage anti-selection, ensure equitable risk rating, detect fraud, and align pricing with risk pool.
- Financial underwriting prevents over-insurance, especially for high sums assured.
- Claims underwriting ensures only valid claims are paid.

Benefits and Drawbacks of Underwriting

- Benefits: reduces anti-selection, ensures fairness, detects fraud, aligns pricing with risk, and manages over-insurance.
- Drawbacks: can create sales barriers, increase expenses, and add complexity (especially for products like income protection).

Outcomes for Substandard Lives

- Possible outcomes: increased premium, reduced benefits, exclusions, deferred decisions, or outright decline.
- Special terms are set based on the degree of extra risk identified.

Alternatives to Medical Underwriting

- Simple health questions, waiting periods, accident-only cover, third-party medicals, and claims underwriting.

- Used to reduce barriers to sales or when medical underwriting is not feasible (e.g., regulatory restrictions).

Underwriting in Group Business

- Free cover limits: automatic acceptance up to a certain benefit level without underwriting.
- Actively-at-work requirements to reduce anti-selection at inception.
- Compulsory cover or minimum take-up rates for voluntary schemes to manage selection risk.
- No financial underwriting needed as salary is known and benefits are formulaic.

Impact of Underwriting Restrictions

- Regulatory bans on medical underwriting lead to community rating, increasing anti-selection risk and average claims cost.
- Insurers may need to adjust pricing, product design, and marketing to attract healthier lives and discourage unhealthy ones.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Practical data checks and reconciliations** — Calculating and interpreting movement reconciliations, lapse/benefit/new business rates, and identifying data inconsistencies or missing items.
- **Purpose and process of underwriting** — Explaining how underwriting manages risk, anti-selection, and ensures fair pricing, including the role of financial and claims underwriting.
- **Impacts of underwriting changes or restrictions** — Analyzing the effects of removing medical underwriting (e.g., due to regulation) on pricing, experience, and business mix, and suggesting mitigation strategies.
- **Comparison of underwriting approaches** — Contrasting stringent vs. simplified underwriting, especially in direct or online sales, and discussing implications for sales, risk, and process.
- **Underwriting for substandard lives** — Outlining possible outcomes and special terms for higher-risk applicants.
- **Group business underwriting** — Describing how underwriting is adapted for group schemes (free cover limits, actively-at-work, compulsory cover).
- **Data validation in reserving and reporting** — Detailing checks needed to ensure data integrity for actuarial valuations and supervisory reporting.

Common Pitfalls & Exam Technique

- Omitting key data checks such as reconciliation with accounting data or failing to mention spot checks and sensitivity tests.
- Not quantifying or interpreting movement rates (lapse, benefit, new business) or failing to compare them to expectations/assumptions.
- Overlooking the impact of underwriting changes on both new and existing business, or not discussing anti-selection risk in detail.

- Confusing financial underwriting (over-insurance) with medical underwriting (health risk).
- Neglecting to mention practical alternatives to medical underwriting when required by regulation or product design.
- Forgetting group-specific underwriting features (free cover limits, actively-at-work) in group business questions.
- Failing to link data checks to their purpose (e.g., ensuring accuracy for reserving, pricing, or regulatory reporting).
- In calculation questions, not showing all steps or not commenting on the reasonableness of results.

#9 Modelling HIGH PRIORITY

24 exam questions | last tested 2025 | recency 12.42

What You Must Know

This topic covers the design, use, and validation of actuarial models in life insurance, focusing on how models are constructed, what they are used for (pricing, reserving, capital, business planning), and how risk and uncertainty are incorporated. It is a frequent exam topic because robust modelling underpins all actuarial advice and regulatory compliance in life insurance.

Core Concepts

Objectives and Requirements of Actuarial Models

- Models are used for financial projections, pricing, reserving, capital assessment, business planning, and risk management.
- Key requirements: validity, adequate documentation, mathematical rigour, and the ability to reflect all material features of the business.

Model Components

- Model points: representative data records for homogeneous policy groups to reduce complexity.
- Parameters: include all variables that could materially affect decisions (e.g., interest rates, mortality, lapses).
- Projection period: frequency must balance reliability and cost-effectiveness.
- Cashflows: include all material real and notional cashflows (premiums, claims, expenses, investment income, reserving movements).
- Dynamic correlations: model interactions between variables (e.g., investment returns and bonus rates).
- Stochastic capability: required for products with guarantees/options; allows simulation of distributions of outcomes.

Profit Calculation and Cashflow Modelling

- Profit is the change in free assets: $Profit = (A_{t+1} - V_{t+1}) - (A_t - V_t)$.

- Distinguish between real cashflows (actual asset movements) and notional cashflows (e.g., reserve increases).
- Investment income on reserves is a real contribution to profit.

Deterministic vs Stochastic Modelling

- Deterministic models use single best-estimate assumptions; sensitivity testing can approximate risk.
- Stochastic models assign probability distributions to key variables, allowing for the assessment of the range and likelihood of outcomes.
- Stochastic models are essential for valuing options/guarantees and quantifying probability of ruin.

Market-Consistent vs Traditional Approaches

- Traditional: uses risk-free rate plus risk margins, assets at book value, assumptions from experience with prudence.
- Market-consistent: assets and liabilities valued at market value, discounting at risk-free rate, explicit risk margins for non-hedgeable risks.

Model Uses: Pricing, Profitability, Solvency

- Pricing: project cashflows for model points, discount at risk discount rate, assess profit criteria (NPV, IRR, DPP), allow for capital requirements.
- Profitability of existing business: use full policy data or model points, project and discount cashflows to calculate PVFP, inform management decisions and embedded/appraisal value.
- Solvency: compare assets and liabilities under regulatory/economic capital rules, use static (point-in-time) or dynamic (multi-year) projections, deterministic or stochastic scenarios.

Sensitivity Testing and Allowance for Risk

- Sensitivity testing: vary assumptions to assess impact on results, identify vulnerabilities, and determine required margins.
- Risk discount rate: includes a risk premium to allow for uncertainty in assumptions.
- Scenario testing: adjust correlated variables together to assess impact of adverse conditions.
- Market-consistent approach: discount at risk-free rate, add explicit risk margins for non-hedgeable risks.

Model Validation and Limitations

- Validation: check model outputs against actual experience, reconcile with published values, and perform reasonableness checks.
- Limitations: model error can arise from data, assumptions, structure, or parameterization; beware of spurious accuracy, especially in stochastic models.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Describing model features for specific products or scenarios** — Detail the required cashflows, states, decrements, and structure (e.g., multi-state models for income protection, cashflow models for complex products).
- **Justifying modelling approaches** — Explain why a cashflow model is needed (complexity, multiple decrements, sensitivity analysis) versus a formula approach.
- **Setting and justifying assumptions** — Discuss sources and rationale for assumptions (experience, market data, prudence, consistency, and market-consistency).
- **Profit testing and pricing process** — Outline the steps in projecting cashflows, discounting, and assessing profitability (NPV, IRR, DPP), including allowance for capital and risk.
- **Market-consistent valuation** — Explain how to value liabilities using market-consistent principles, including discounting at risk-free rates and adding risk margins.
- **Model points selection** — Describe how to group policies for modelling, especially when full data is impractical.
- **Sensitivity and scenario testing** — Show how to test the impact of changes in assumptions or scenarios on model outputs and profitability.
- **Risks and limitations of models** — Identify sources of model error, limitations, and the importance of validation and documentation.
- **Equity and fairness in unit pricing** — Discuss how unit price calculations can affect fairness between policyholder generations.

Common Pitfalls & Exam Technique

- Omitting key cashflows (e.g., reinsurance, expenses, reserving movements) in model descriptions.
- Failing to distinguish between real and notional cashflows, especially in profit calculations.
- Not specifying the need for multi-state or stochastic models when multiple decrements or guarantees are present.
- Overlooking the need for sensitivity/scenario testing or not explaining its purpose clearly.
- Confusing market-consistent and traditional valuation approaches; always state the discount rate and asset/liability valuation basis.
- Neglecting to mention model validation, limitations, and the importance of documentation.
- In pricing/profit testing, forgetting to state the use of model points or the need to gross up results for the full portfolio.
- Not addressing how assumptions are set or justified, especially for new or unique products.
- In questions on unit pricing or equity, failing to discuss how timing of charges or investment returns can affect fairness between policyholders.

#10 Guarantees and options MEDIUM PRIORITY

24 exam questions | last tested 2025 | recency 9.37

What You Must Know

This topic covers the methods for determining the cost of guarantees and options in life insurance products, focusing on both investment guarantees (such as minimum maturity or annuity guarantees) and mortality options (such as guaranteed insurability). It is frequently examined because these features introduce significant risk and complexity to product design, pricing, reserving, and capital management.

Core Concepts

Implications of Offering Investment Guarantees

- Transfers investment risk from policyholder to insurer, exposing the insurer to potential asset shortfalls at claim time.
- Influences investment strategy: matching the guarantee limits upside, while riskier assets may increase expected returns but also risk.
- Guarantees exercisable at any time (e.g., surrender guarantees) are harder to manage than fixed maturity guarantees due to timing uncertainty.
- Pricing requires sophisticated modelling, often stochastic, to estimate the additional premium needed.
- Guarantees increase supervisory reserve and solvency capital requirements, causing higher initial capital strain.

Methods for Valuing Investment Guarantees

- Option-pricing techniques: Treat guarantees as financial options (e.g., puts or calls) and use market-consistent valuation or formulas like Black-Scholes.
- Stochastic simulation: Simulate future asset returns and policyholder behavior to estimate the expected cost of the guarantee.

Option Pricing Techniques

- Guarantees are valued as options: maturity guarantees as European puts, surrender guarantees as American puts, guaranteed annuity rates as calls on bonds or swaps.
- If a replicating market-traded option exists, its price is used (market-to-market approach).
- Option pricing formulas (e.g., Black-Scholes) provide efficient, market-consistent values and avoid nested simulations.

Stochastic Simulation for Investment Guarantees

- Simulate future asset values using a chosen probability distribution for returns.
- Model policyholder behavior, especially when options are likely to be exercised (e.g., when guarantees are "in the money").
- For each simulation, calculate the shortfall (if any) between the fund value and the guarantee; discount these costs to present value.
- The expected cost is the average across all simulations.
- Method can be adapted for different guarantee types (e.g., surrender, maturity, annuity rate guarantees).

Implications of Offering Mortality Options

- Anti-selection risk: Policyholders in poorer health are more likely to exercise options (e.g., to increase cover without underwriting).
- Financial cost: The cost is the difference between the standard premium and the premium that would be charged under full underwriting, multiplied by the proportion and health of those exercising.
- Higher capital strain: Reserves must allow for higher expected mortality and uncertainty in take-up rates.

Factors Affecting the Risk of Mortality Options

- Take-up rates: Proportion of eligible policyholders expected to exercise the option.
- Mortality of exercisers: Typically higher than standard due to anti-selection.
- Mortality of non-exercisers: May be lower than standard to balance overall experience.
- Benefit choice: If multiple benefit levels are available, assume the most costly for the insurer.
- Additional expenses: Notification and administration costs for option exercise.

Valuing Mortality Options

- Project cashflows allowing for survival to option date, probability of exercise, and expected mortality of exercisers.
- Cost is the present value of extra cover at standard rates, less the premium charged for the option.
- Assumptions needed: take-up rates (100% for conventional, experience-based for realistic), mortality rates for exercisers/non-exercisers, benefit choice, and expenses.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Defining and explaining the cost of mortality options** — Clearly state the cost as the excess premium required due to anti-selection and the factors influencing it (proportion exercising, health status).
- **Key factors in pricing mortality options** — Discuss take-up rates, expected mortality of exercisers, anti-selection, expenses, and reserving implications.
- **Manifestations and mitigation of anti-selection** — Identify how anti-selection arises (e.g., unhealthy lives exercising, healthy lives not exercising) and propose controls (e.g., restrictions, premium loadings, limits on cover).
- **Valuation methods for investment guarantees** — Describe and compare option-pricing and stochastic simulation approaches, including their application to specific product features (e.g., guaranteed maturity, surrender, annuity rates).
- **Impact of guarantees on investment strategy and reserving** — Explain how adding guarantees changes asset selection, risk management, and capital requirements.

- **Assumptions for supervisory reserves** — Set out prudent assumptions for mortality, expenses, and inflation, especially for products with guarantees.
- **Cashflow projections for options** — Describe the expected cashflows under products with options, including premiums, claims, expenses, and option exercise.
- **Product design and risk management** — Suggest ways to manage or limit the risks associated with guarantees and options (e.g., product restrictions, reinsurance, reserving).

Common Pitfalls & Exam Technique

- Failing to distinguish between the cost of the option and the cost of the underlying cover — always specify the incremental cost due to the option.
- Ignoring anti-selection or assuming standard mortality for exercisers — always allow for higher mortality in those exercising options.
- Assuming 100% take-up without justification — state whether you are using a conventional (100%) or realistic (experience-based) approach, and justify your choice.
- Not discussing the impact on capital and reserving — always mention increased capital strain and reserving needs for guarantees/options.
- Overlooking expenses and administrative costs associated with options — include these in pricing and reserving.
- Describing only one valuation method for investment guarantees — always mention both option-pricing and stochastic simulation, and when each is appropriate.
- Not addressing policyholder behavior in stochastic models — explicitly state how exercise behavior is modelled.
- Forgetting to discuss mitigation strategies for anti-selection — always suggest practical controls (e.g., limits, waiting periods, premium loadings).

#11 Investments MEDIUM PRIORITY

23 exam questions | last tested 2024 | recency 9.33

What You Must Know

This topic covers how life insurers select, manage, and justify their investment strategies, focusing on matching assets to liabilities, maximizing returns, and managing risk. It is frequently examined because investment decisions directly impact solvency, policyholder expectations, and regulatory compliance.

Core Concepts

Asset Characteristics

- Key characteristics: security, yield/return, spread, term, expenses, marketability, tax, and currency.
- Return: consider expected return, real vs. nominal, income vs. capital, running yield, variance, and tax impact.

- Security: assess risk of default and market value volatility; higher returns often mean higher risk.
- Marketability: liquidity and dealing costs vary by asset (e.g., government bonds vs. property).
- Term: match asset duration to liability duration to manage interest rate risk.
- Currency: assets should match the currency of liabilities to avoid currency risk.

Principles of Investment

- Matching: select assets to match the nature, term, and currency of liabilities to minimize risk.
- Maximizing return: seek to maximize total return (income + capital gains) within acceptable risk limits.
- Free assets: the level of free assets determines how much the company can deviate from strict matching to pursue higher returns.
- Perfect matching is usually not desirable unless there are no free assets, regulatory constraints, or fixed/certain liabilities.

Asset-Liability Matching

- Liabilities may be: fixed in money terms (match with bonds/cash), index-linked (match with index-linked bonds, property, or equities), discretionary (back with real assets), or investment-linked (hold exact matching assets).
- Duration matching: align asset and liability discounted mean terms to reduce sensitivity to interest rate changes.
- Equity-indexed liabilities: require index replication, which can be complex and costly due to rebalancing and transaction costs.

Free Assets and Investment Strategy

- Free assets provide a buffer, allowing for riskier investments and potential for higher returns.
- Greater free assets = more freedom to mismatch assets and liabilities, within risk appetite and regulatory limits.
- Mismatching is rare for investment-linked liabilities due to regulation, risk transfer, and administrative simplicity.

Developing an Investment Strategy

- Step 1: Match assets to the nature, term, and currency of liabilities (investment-linked, index-linked, fixed guarantees, discretionary benefits).
- Step 2: Assess free assets and determine how much risk can be taken for higher returns (e.g., equities, property, overseas assets).
- Ensure sufficient liquidity for operational needs.
- Use model office simulations to test strategies and asset-liability interactions over time.

Stochastic Asset-Liability Modelling

- Used to value guarantees and test investment strategies under uncertainty.
- Risks: parameter sensitivity, computational constraints, and lack of dynamic management actions can lead to misleading results.

Regulatory and Legislative Constraints

- Regulations may impose limits on asset classes, individual exposures, currency, duration, credit rating, and require mismatching reserves or liquidity buffers.
- Legislation can restrict investment freedom and require specific matching or minimum holdings.

Bonus Structures in With-Profits Business

- Types: regular reversionary (simple, compound, super-compound), special reversionary, and terminal bonuses.
- Bonus strategy affects investment approach: higher terminal bonuses allow more investment risk; higher reversionary bonuses require more secure, predictable returns.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Principles of investment strategy** — Explain matching (nature, term, currency), maximizing return, and the role of free assets in determining risk appetite.
- **Asset-liability matching for specific products** — Recommend suitable asset portfolios for different policy types (e.g., with-profits, without-profits, annuities, unit-linked) and justify choices.
- **Impact of regulation and legislation** — Outline how legal and regulatory constraints affect investment options and strategy implementation.
- **Bonus structure and investment implications** — Discuss how the mix of reversionary and terminal bonuses influences asset selection and risk-taking.
- **Valuing guarantees and stochastic modelling** — Describe methods (e.g., stochastic simulation, option pricing) for valuing investment guarantees and discuss modelling limitations.
- **Responding to market conditions** — Analyze how changes in investment markets (e.g., volatility, downturns) affect the insurer's asset-liability position and bonus declarations.
- **Regulatory examples** — List and explain types of investment regulations imposed on insurers.
- **Developing and justifying investment strategies** — Apply the stepwise approach: match liabilities, assess free assets, and adapt for operational and regulatory needs.

Common Pitfalls & Exam Technique

- Omitting discussion of all three matching dimensions: nature, term, and currency.
- Failing to link asset choice directly to liability characteristics (e.g., using equities for fixed liabilities).
- Ignoring the impact of free assets on risk appetite and investment freedom.
- Overlooking regulatory or legislative constraints that may override optimal investment choices.

- Not distinguishing between types of bonuses and their implications for investment risk.
- Neglecting to mention liquidity needs for operational cash flow.
- Describing stochastic modelling without noting parameter sensitivity or practical limitations.
- Forgetting to address both income and capital components of return, and the effect of tax.
- Not providing practical examples or justifications for asset selection in exam scenarios.

#12 Reinsurance MEDIUM PRIORITY

22 exam questions | last tested 2025 | recency 8.98

What You Must Know

This topic covers the role of reinsurance in managing risk for life insurers, including the main types of reinsurance contracts, their structures, purposes, and the strategic considerations in their use. Reinsurance is a frequent exam focus because it is central to risk management, capital efficiency, and product design in life insurance.

Core Concepts

Main Types of Reinsurance Contracts

- **Original Terms (Coinsurance):** Proportional sharing of all aspects of the insurance contract; insurer sets the premium, reinsurer pays a proportion and receives a commission.
- **Risk Premium:** Reinsurer sets its own premium rates for a specific portion of risk, often as recurring single premiums; aligns well with unit-linked business.
- **Level Risk Premium:** Reinsurer sets a level premium for its share of risk; insurer calculates its own premium rates accordingly.
- **Excess of Loss (XL):** Non-proportional; reinsurer pays for losses above a retention limit. Includes catastrophe, stop loss, and risk-based layers.
- **Financial Reinsurance (Fin Re):** Used for capital management and solvency improvement, not primarily for risk transfer; includes financing arrangements and contingent loans.
- **Facultative vs Obligatory:** Facultative allows parties to choose which risks to cede/accept; obligatory requires ceding/accepting as per treaty terms.

Allocation Methods for Reinsured Amount

- **Quota Share:** Fixed percentage of each policy ceded to the reinsurer.
- **Individual Surplus:** Reinsurer covers the excess of policy benefit over the insurer’s retention limit.

Key Features and Suitability of Reinsurance Types

- **Original Terms:** Proportional sharing, significant commission, reduces new business strain; less suitable for with-profits or unit-linked business.
- **Risk Premium:** Age-rated, recurring premiums, aligns with unit-linked mortality charges, minimal commission.
- **Excess of Loss:** Non-proportional, unknown claim amount in advance, upper cover limits, suitable for capping large/aggregate losses.
- **Financial Reinsurance:** Improves solvency via upfront cash, but little impact on embedded value; effectiveness depends on regulatory regime.

Reasons for Reinsuring

- **Capital Management:** Reduce required capital or leverage reinsurer's capital.
- **Parameter Risk Reduction:** Share risk of mis-estimated assumptions; access reinsurer's technical expertise.
- **Claims Fluctuation Reduction:** Cap exposure to large individual or aggregate claims.
- **New Business Strain Relief:** Reinsurance commissions/loans and reduced reserving requirements.
- **Technical Assistance:** Underwriting, product design, pricing, and systems support.
- **Cost Reduction:** Lower risk margins, regulatory/tax arbitrage, capital efficiency.

Considerations Before Reinsuring

- **Type of Reinsurance:** Chosen based on business needs, cost, and product type.
- **Cost of Reinsurance:** Reduces expected profit, may impose stricter risk management, increases counterparty risk.
- **Retention Limits:** Set based on risk appetite, free assets, and portfolio distribution; higher free assets allow higher retention.
- **Alternatives:** Mortality fluctuation reserves may be considered, but have limitations.
- **Legal and Counterparty Risks:** Ensure treaties are robust and reinsurer is financially strong.

Setting Retention Limits and Stochastic Simulation

- **Retention Limit:** Maximum risk retained per policy; excess ceded to reinsurer.
- **Factors (P.R.O.F.E.R B.C.):** Product features, Risk appetite, Operational capacity, Free assets, Experience, Reinsurance terms, Business concentration, Company size.
- **Stochastic Simulation:** Model total costs (reinsurance + reserve financing) for different retention limits to optimize for a target ruin probability.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Suitability of Reinsurance Types for Product Designs:** Justify which reinsurance structures (e.g., risk premium, XL, quota share) are appropriate for specific products (e.g., unit-linked, income protection) and why.
- **Risk Mitigation via Reinsurance:** Explain how reinsurance can be used to manage mortality, morbidity, claims volatility, and new business strain for new or existing products.
- **Retention Limits and Their Determination:** Define retention limits, list factors influencing their setting, and describe how to use stochastic simulation to optimize retention and reserve strategies (including formulas).
- **Trade-offs and Downsides:** Discuss the costs, operational impacts, and counterparty risks of reinsurance versus alternatives like fluctuation reserves.
- **Technical and Strategic Considerations:** Identify when facultative vs obligatory arrangements are preferable, and the strategic advantages of each.
- **Reinsurance in Product Development:** Apply reinsurance concepts to new product launches, especially in unfamiliar markets or with innovative features.
- **Commission and Profit-Sharing:** Factors affecting reinsurance commission levels and how profit-sharing arrangements work.

Common Pitfalls & Exam Technique

- Omitting non-proportional (XL, stop loss) or financial reinsurance when asked for all types relevant to a scenario.
- Failing to link the choice of reinsurance type to the specific product features or risks (e.g., using OTR for unit-linked without noting its limitations).
- Not addressing both the advantages and disadvantages (including cost, operational, and counterparty risks) in discussion questions.
- Neglecting to mention technical assistance and expertise as a benefit of reinsurance.
- Forgetting to discuss the impact of reinsurance on capital requirements, solvency, and new business strain.
- Not providing formulas or a clear stepwise approach when asked about stochastic simulation or optimization of retention limits.
- Overlooking the need to consider legal and regulatory factors, especially for financial reinsurance or cross-border arrangements.
- In calculation or modeling questions, not clearly stating assumptions or the rationale for chosen parameters.

#13 Reserving MEDIUM PRIORITY

22 exam questions | last tested 2022 | recency 8.40

What You Must Know

This topic covers the principles, methods, and regulatory requirements for setting reserves and solvency capital for life insurance companies. It is frequently examined because reserving underpins solvency, financial reporting, and risk

management, and requires understanding both technical calculations and regulatory context.

Core Concepts

Purpose and Methods of Reserving

- Reserves demonstrate solvency to regulators and provide a realistic financial position for management.
- Primary reserving methods: Gross Premium Valuation (GPV) and Net Premium Valuation (NPV).

Gross Premium Valuation (GPV)

- Explicitly values all future policyholder liabilities: benefits, expenses, bonuses, guarantees, and options, less future office premiums.
- Assumptions include mortality, lapse, expenses, and discount rates (often market-consistent).
- Formula: Reserve = EPV(Benefits) + EPV(Expenses) - EPV(Premiums).
- Can produce negative reserves on a realistic basis, but negative reserves are zeroised for solvency.
- Highly sensitive to changes in valuation basis; profit emerges gradually over time.

Net Premium Valuation (NPV)

- Reserves = EPV(Future benefit outgo) - EPV(Future net premiums).
- Net premium is a theoretical premium covering only guaranteed benefits, using current valuation assumptions for interest and mortality.
- No explicit allowance for expenses or bonuses; relies on implicit margins.
- Simple, low data requirements, but not suitable for single premium business or explicit profit recognition.

Non-Unit Reserves (Unit-Linked Business)

- Non-unit reserve = EPV[Non-unit Outgo] - EPV[Non-unit Income].
- Outgo: future costs (expenses, excess benefits); Income: unallocated premiums, charges, bid-offer spreads.
- Calculated via cashflow projection, working backwards from policy end to valuation date.
- Prudential valuation: non-unit reserve set to zero if discounted cashflow is non-negative; best estimate valuation allows negative non-unit reserves.

Negative Non-Unit Reserves

- Arise when future positive cashflows (charges) exceed future outgo (expenses/benefits).
- Effectively treat the contract as an asset, reducing initial capital strain.
- Regulatory conditions: must not breach surrender value floor, no future negative cashflows, aggregate non-unit reserves must not be negative, and regulatory permission is required.
- Negative reserve is a "loan" repaid by future profits; must avoid future valuation strain.

Active vs Passive Valuation Approaches

- Passive: infrequent updates, locked-in assumptions, assets at book value, simple solvency formulas.
- Active: frequent updates, market-consistent assumptions, assets/liabilities at market value, risk-based capital (e.g., VaR).

Market-Consistent Valuation (MCV)

- Assets and liabilities valued at fair value; liabilities discounted at risk-free rates (from government bonds or swaps).
- Illiquidity premium may be added for long-term predictable liabilities with matching assets.
- Non-market risks valued using best estimate assumptions plus explicit risk margin (cost of capital approach).
- Requires risk-neutral calibration of stochastic models to replicate market prices.

Cost of Capital Approach

- Projects future required capital (SCR), multiplies by cost of capital rate, and discounts to present value to determine risk margin.
- Cost of capital rate reflects incremental and frictional costs of holding capital above risk-free rate.
- Used in market-consistent valuation, embedded value, and asset share calculations.
- Simplified drivers (e.g., projected reserves) may be used to approximate future capital requirements.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Regulatory and prudential requirements for negative non-unit reserves** — State and explain conditions (e.g., surrender value floor, aggregate limits, repayment of "loan", prudence of assumptions) for holding negative non-unit reserves.
- **Calculation and rationale for unit and non-unit reserves in unit-linked contracts** — Describe components, cashflow projections, and when non-unit reserves are needed.
- **Setting prudent assumptions for supervisory reserves** — Explain how to set mortality, expense, withdrawal, and inflation assumptions for different products under a prudent (supervisory) basis.
- **Comparison of active vs passive valuation approaches** — Define, compare, and discuss implications for solvency, profit emergence, and management information.
- **Market-consistent valuation (MCV) in practice** — Outline steps for calculating MCV of liabilities, including use of risk-free rates, illiquidity premium, and risk margin.
- **Risk-neutral calibration and cost of capital** — Describe calibration process, application of cost of capital approach, and calculation of risk margin.

- **Data and consistency checks for reserving** — List types of accounting and fund management data required, and outline data checks for consistency between policy, accounting, and fund data.
- **Analysis of surplus and management actions** — State reasons for surplus analysis and possible management responses to adverse results (e.g., expense overruns).

Common Pitfalls & Exam Technique

- Omitting regulatory conditions or aggregate limits when discussing negative non-unit reserves.
- Failing to distinguish between unit and non-unit reserves, or not explaining why each is needed.
- Not specifying the use of prudent (not best estimate) assumptions for supervisory reserves.
- Confusing GPV and NPV methods, especially regarding treatment of expenses and bonuses.
- Neglecting to mention the surrender value floor when calculating reserves for unit-linked contracts.
- Overlooking the need for explicit risk margins and cost of capital in market-consistent valuation.
- Not providing practical steps or examples when asked about data checks or surplus analysis.
- Writing generic answers on valuation approaches without linking to solvency, profit emergence, or management use.

#14 External environment MEDIUM PRIORITY

20 exam questions | last tested 2023 | recency 9.26

What You Must Know

This topic covers the wide range of external environmental factors that influence the risks, profitability, and operational decisions of life insurers. It is a frequent exam focus because it tests your ability to identify, explain, and apply these factors to real-world scenarios, demonstrating both breadth and depth of understanding.

Core Concepts

Volume and Mix of New Business

- Expected new business volume affects expense recovery, profitability, solvency strain, and operational capacity.
- Low volumes may not cover fixed costs; high volumes can cause new business strain and capital pressure.
- Business mix (by age, gender, sum assured, channel, etc.) impacts risk profile, capital needs, claims, and profitability.

Target Market and Customer Demand

- Assess if there is sufficient demand and product understanding in the target market.
- Consider affordability, financial literacy, cultural attitudes, and policyholder reasonable expectations (PRE).
- Poor understanding or unmet needs can lead to low sales, mis-selling, complaints, and reputational/regulatory risk.

Distribution Channel and Distributor Behaviour

- Choice of channel (independent intermediaries, tied agents, own salesforce, direct marketing) affects risk, persistency, and anti-selection.
- Commission structures and sales incentives can create conflicts of interest, mis-selling, or policy churning.
- Monitor persistency, complaints, and claims by channel; ensure adequate training and disclosure.

Competition

- Competition on price, benefits, underwriting, service, and distribution can reduce margins and increase lapse risk.
- Competitor actions may force product redesign or repricing.

Economic Environment

- Inflation, interest rates, investment returns, unemployment, and economic growth affect affordability, lapses, expenses, and profitability.
- Volatile markets increase capital requirements and may reduce demand for insurance products.

Legal and Regulatory Environment

- Legal constraints limit insurer discretion (e.g., PRE, contract law, unfair clauses).
- Regulation may restrict product design, pricing, terms, distribution, reserving, capital, investments, and disclosure.
- Regulatory changes can increase costs, require product redesign, or additional capital.

Tax and Fiscal Environment

- Tax treatment of premiums, benefits, investment income, and profits affects product attractiveness and profitability.
- Tax changes can impact demand, pricing, and policyholder behaviour; long-term contracts are exposed to fiscal risk.
- Tax rules may impose product design constraints (e.g., minimum cover for tax-advantaged status).

Expenses and Operational Capability

- Insurer must have adequate systems, staff, and expertise to administer products efficiently.
- Expenses split into acquisition, renewal, and claim; fixed vs variable costs create volume risk.

- Expense inflation, especially earnings inflation, is a key risk for long-term guarantees.

Claims Environment

- Expected claims experience, anti-selection, fraud, and clarity of definitions affect profitability and risk.
- Poor claims management increases costs, disputes, and reputational risk.

Policyholder Behaviour

- Lapses, surrenders, premium holidays, and option take-up materially affect profitability, reserves, and capital.
- Anti-selective behaviour (keeping policies when valuable, lapsing when not) worsens experience.

Reinsurance Availability

- Reinsurance can reduce volatility, manage large claims, support new products, and reduce capital needs.
- Introduces counterparty risk and may reduce retained profits; cost and terms affect product viability.

Capital and Solvency Requirements

- Capital needs depend on guarantees, long-term liabilities, assumption uncertainty, and risk exposures.
- High capital requirements may require higher premiums or limit product viability; rapid growth can strain solvency.

Reputational and Conduct Risk

- Mis-selling, unclear terms, poor claims handling, and unfair charges can damage reputation and attract regulatory scrutiny.
- Reputational damage reduces sales and harms relationships with distributors and policyholders.

Professional Guidance

- Professional standards constrain actuarial and management discretion, protecting policyholders and ensuring ethical conduct.
- Guidance supports capital oversight, ethical distribution, and consumer confidence, enabling market growth and product flexibility.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Application of external factors to new markets or products** — E.g., expanding to new target markets (blue-collar vs white-collar), launching new products, or entering new distribution partnerships. Answers must tailor external environment factors to the specific scenario.
- **Regulatory and legal restrictions** — Identify and explain types of regulatory constraints (on product design, pricing, reserving, investments, disclosure, etc.) and their practical impact.

- **Economic environment impacts** — Discuss how changes in inflation, interest rates, and economic conditions affect sales, profitability, and product attractiveness.
- **Distribution channel risks and pricing implications** — Explain how different channels affect risk profile, anti-selection, persistency, expenses, and pricing assumptions.
- **Reinsurance considerations** — Outline when and why reinsurance is needed, types of reinsurance, and how it supports risk management for new or risky business.
- **Claims and underwriting practices** — Compare group vs individual underwriting, discuss anti-selection, and suggest ways to manage claims and underwriting risk.
- **Expense and commission structure** — Distinguish between fixed and variable costs, explain new business strain, and discuss the impact of expense overruns or inflation.
- **Policyholder behaviour and conduct risk** — Assess the impact of lapses, mis-selling, and reputational risk on profitability and future business.

Common Pitfalls & Exam Technique

- Listing factors without explaining their relevance or impact on the insurer's risk, profitability, or operations.
- Failing to tailor answers to the specific scenario (e.g., not adjusting for a new target market or product type).
- Overlooking the interaction between factors (e.g., how economic conditions affect both policyholder behaviour and capital requirements).
- Ignoring the practical implications of regulatory, tax, or professional guidance changes.
- Not distinguishing between fixed and variable expenses, or between different distribution channels' risk profiles.
- Neglecting to mention reinsurance as a risk management tool when discussing new or high-risk business.
- Missing the importance of monitoring, feedback, and the actuarial control cycle in managing external environment risks.
- For high-mark questions, not providing enough breadth (distinct points) or depth (explanation) — use headings and aim for at least 2 points per mark.

#15 Product design LOW PRIORITY

17 exam questions | last tested 2019 | recency 5.38

What You Must Know

This topic covers the principles, practicalities, and trade-offs involved in designing life insurance products. It is a frequent exam focus because product design decisions directly affect profitability, risk, marketability, regulatory compliance, and operational efficiency.

Core Concepts

Reasons for Reviewing or Designing New Products

- Identifying gaps in the product range or responding to market demand
- Adapting to legislative, fiscal, or regulatory changes
- Addressing inadequacies or loss-making products
- Maintaining competitiveness and marketability
- Improving risk management and capital efficiency

Key Factors in Product Design (CAMP SPRING)

- Customer needs: Ensuring the product meets specific financial needs
- Profitability: Covering claims, expenses, and providing adequate return
- Marketability: Attractiveness, affordability, and alignment with customer expectations
- Competitiveness: Differentiation, pricing, and distribution channel considerations
- Financing: Managing capital strain, reserving, and funding requirements
- Risk characteristics: Acceptable risk profile, risk transfer, and margins
- Guarantees and sensitivity: Impact of guarantees on capital and profit volatility
- Cross-subsidies: Managing expense allocation and fairness
- Administration systems: System constraints on features and charges
- Consistency: Alignment with existing products for operational and marketing reasons
- Regulatory constraints: Compliance with legal and supervisory requirements

Profitability and Its Constraints

- Profit must cover claims, expenses, and provide for shareholder returns
- Profitability is sensitive to volumes, lapses, expenses, and investment returns
- Constraints include competition, cross-subsidies, and regulatory profit caps
- Profit criteria: NPV, IRR, and discounted payback period (DPP)

Marketability and Competitiveness

- Product must be attractive, affordable, and meet customer expectations
- Distribution channels influence design complexity and price sensitivity
- Competitiveness may require innovation or product differentiation
- Profit-volume trade-off: Lower per-policy profit may be offset by higher sales volume

Financing and Capital Management

- Designs should minimize initial capital strain and optimize use of free assets
- Unit-linked products generally require less capital than non-linked
- Guarantees and onerous features increase reserving and capital requirements

- Charging structures and premium frequency affect capital efficiency

Risk Management in Product Design

- Risk can be managed by reviewable terms, risk transfer (reinsurance), or product features
- Risk margins and rider benefits can limit exposure
- Designs should consider the company's risk appetite and ability to absorb losses

Guarantees and Sensitivity of Profit

- Guarantees increase capital strain and profit sensitivity to adverse experience
- Onerousness depends on size, duration, reserves, and likelihood of guarantee being exercised
- Profit sensitivity is driven by investment return, mortality/morbidity, expenses, and withdrawals
- Designs can reduce sensitivity by limiting guarantees, using reviewable charges, and matching income/outgo

Cross-Subsidies and Mutuality

- Cross-subsidy: Sharing of non-risk elements (mainly expenses) between policyholders
- Mutuality: Pooling of insurance risks (e.g., mortality) across policyholders
- Simple charging structures often create cross-subsidies, especially by policy size
- Trade-off between simplicity (marketability) and equity (fairness)

Administration Systems

- System capabilities may limit product features or charging structures
- Consistency with existing systems reduces costs and operational risk

Consistency with Other Products

- Consistency ensures operational efficiency and policyholder equity
- Prevents marketing risk from perceived unfairness by existing policyholders
- Facilitates ease of transition and brand alignment

Interaction and Trade-offs Between Design Factors

- Profitability/risk control often conflicts with marketability/competitiveness
- Rigorous underwriting may reduce risk but deter sales
- Simplicity may create cross-subsidies; innovation may challenge consistency
- Distribution channel dictates product complexity and features

Regulatory Constraints

- Regulations protect policyholders and ensure insurer solvency
- May restrict underwriting, pricing, guarantees, and product features

- Industry agreements (e.g., no medicals, no HIV exclusions) can increase risk and require design mitigation

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Application of product design factors to specific scenarios** — E.g., how to design for low-income markets, or when adding new features/options (2016 JUNE Q4, Q8; 2017 JUNE Q6, Q8)
- **Analysis of trade-offs and interactions** — E.g., balancing profitability, risk, and marketability; impact of cross-subsidies; effect of guarantees on capital and profit sensitivity (2016 NOV Q5; 2019 NOV Q2, Q8)
- **Impact of regulatory and market constraints** — E.g., how industry agreements or regulatory changes affect design, underwriting, and risk (2016 JUNE Q8; 2017 JUNE Q6)
- **Designing for specific customer needs and distribution channels** — E.g., differences between group and individual products, or direct vs. bank distribution (2017 NOV Q5; 2016 JUNE Q8)
- **Profitability assessment and sensitivity analysis** — E.g., how profit is affected by business mix, lapses, expenses, and guarantees; how to reduce profit sensitivity (2016 NOV Q5; 2019 NOV Q2)
- **Specification of product features** — E.g., benefit structure, guarantees, options, expense charges, and eligibility (2019 NOV Q2; 2017 JUNE Q8)
- **Consideration of operational and administrative constraints** — E.g., system limitations, consistency with existing products (2017 NOV Q5; 2016 JUNE Q4)

Common Pitfalls & Exam Technique

- Omitting discussion of trade-offs or interactions between design factors (e.g., risk vs. marketability)
- Focusing only on profitability or customer needs without addressing regulatory, operational, or capital constraints
- Neglecting the impact of distribution channels and administration systems on feasible product features
- Failing to address cross-subsidies and their implications for fairness and marketability
- Not considering how guarantees and options affect capital requirements and profit sensitivity
- Ignoring regulatory or industry-imposed constraints (e.g., underwriting restrictions, pricing limits)
- Not tailoring answers to the specific product type or market segment in the question
- Forgetting to specify key product features (benefits, charges, eligibility, options) when asked for a product design outline
- Overlooking the need for consistency with existing products and operational processes
- In scenario questions, always relate general principles back to the specifics given (e.g., low-income market, group vs. individual, unit-linked vs. non-linked)

#16 Surrender values LOW PRIORITY

14 exam questions | last tested 2025 | recency 5.69

What You Must Know

This topic covers the principles, methods, and practical considerations for determining surrender values for conventional without-profits life insurance contracts. It is a frequent exam focus because surrender values directly affect policyholder fairness, insurer profitability, and regulatory compliance.

Core Concepts

Principles for Determining Surrender Values

- Policyholder Reasonable Expectations (PRE): Surrender values should align with what policyholders have been led to expect, based on past practice, marketing, and policy documents.
- Fairness: Maintain equity between policyholders who surrender and those who remain, and between policyholders and shareholders.
- Asset Share Constraint: On average, surrender values should not exceed the earned asset share to avoid insurer losses.
- Consistency: Surrender values should progress smoothly over time and be consistent with projected maturity values at later durations.
- Early Duration Considerations: Avoid surrender values that appear too low relative to premiums paid, but recognize that initial expenses may make early asset shares negative.
- Regulatory Compliance: Surrender value methods must comply with local regulations and be responsive to market conditions.
- Competition: Surrender values should be competitive with those offered by other insurers.
- Stability: Avoid frequent or unjustified changes in surrender value methodology.
- Calculation and Communication: Methods should be simple to calculate, clearly documented, and easy to explain to policyholders.
- Selection Risk: Design should minimize the risk of anti-selection, including lapse and re-entry risks.

Methods of Calculating Surrender Values

- Retrospective Method: Based on the asset share or accumulated net cashflows, reflecting actual experience to date. Used at early durations to avoid unreasonably low values.
- Prospective Method: Present value of future benefits and expenses, net of future premiums, using best estimate assumptions. Ensures smooth progression to maturity value and is sensitive to future assumptions.
- Blended Method: Uses retrospective values at early durations and transitions to prospective values later, balancing expense recovery and consistency with maturity values.
- Unit-Linked Method: Surrender value equals the bid value of allocated units less any surrender penalty, typically used for unit-linked business.

Profit Retention on Surrender

- Insurers retain profit on surrender to meet original profit requirements, prevent lapse-and-re-entry, and maintain equity among policyholders.
- Profit on surrender is the difference between asset share and the surrender value paid.
- Profit can be split into accrued profit (to date) and capitalized future profit (from early termination).
- The choice of surrender basis (best estimate, premium, or blended) determines how much future profit is capitalized and retained by the insurer.

Determining the Surrender Value Basis

- Retrospective Basis: Uses actual past experience for investment returns, expenses, mortality, and tax, possibly adjusted for profit retention or marketability.
- Prospective Basis: Sets assumptions for future interest rates (typically based on current yields of matching assets), expenses (from latest investigations), mortality (may reflect lighter rates for leavers), and inflation (consistent with investment returns). Excludes prudential margins.

Suitable Surrender Value Path

- No value at inception, minimum of premiums paid at early durations, increasing in line with asset share at intermediate durations, and converging to maturity value at later durations.
- Profit retention increases as maturity approaches, so surrendering is as profitable as a policy reaching maturity.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Stating and applying surrender value principles** — List and explain the key principles (PRE, fairness, asset share, consistency, etc.) and apply them to new product designs or proposals.
- **Describing and comparing calculation methods** — Explain retrospective, prospective, blended, and unit-linked methods, including when and why each is used.
- **Discussing profit extraction and basis choice** — Analyze how profit is retained on surrender, the impact of different bases (premium, best estimate, blended), and how this affects fairness and profitability.
- **Determining and justifying basis assumptions** — Detail how interest, expenses, mortality, and inflation assumptions are set for both retrospective and prospective methods, referencing asset backing and experience investigations.
- **Evaluating the impact of external events** — Assess how changes in interest rates, mortality, or other factors affect asset share, surrender value, and retained profit.
- **Critiquing surrender value proposals** — Discuss the suitability of specific surrender value designs, especially regarding early duration values, negative policy values, and regulatory or competitive context.

- **Unit-linked policy considerations** — Outline how surrender values and penalties are set for unit-linked products, and discuss the fairness and adequacy of proposed penalty structures.
- **Suggesting ways to increase profit on surrenders** — Propose and justify changes to surrender value approaches to improve insurer profitability, such as adjusting bases or penalties.

Common Pitfalls & Exam Technique

- Omitting key principles such as PRE, fairness, or asset share constraints in principle-based questions.
- Failing to distinguish between retrospective and prospective methods, or not explaining when each is appropriate.
- Not addressing both early and late duration considerations, especially the need for smooth progression to maturity value.
- Ignoring the impact of assumption choice (interest, expenses, mortality) on the calculated surrender value and retained profit.
- Overlooking the need to justify basis selection with reference to actual experience or asset backing.
- Neglecting to discuss the implications of negative asset shares or policy values at early durations.
- Not considering regulatory, competitive, or communication aspects when critiquing surrender value proposals.
- In calculation or discussion questions, always state the method, the basis, and the rationale for each assumption used.
- For unit-linked products, clearly explain the penalty structure and its fairness, not just the formula.

#17 Alterations LOW PRIORITY

11 exam questions | last tested 2020 | recency 3.43

What You Must Know

This topic covers the principles, methods, and practical considerations for determining alteration terms for without-profits life insurance contracts, especially paid-up values and other policy changes. It is frequently examined because alterations affect fairness, profitability, and policyholder expectations, and require careful actuarial judgment to ensure consistency and avoid anti-selection.

Core Concepts

Paid-Up Contracts

- A paid-up contract arises when a policyholder stops paying premiums but retains a reduced sum assured, rather than surrendering for cash.
- The paid-up sum assured should be calculated to reflect the policy's earned asset share and be consistent with surrender values and maturity values.

- Ongoing administration expenses and possible higher mortality assumptions (due to anti-selection) must be considered.

Principles for Setting Alteration Terms

- Affordability: New terms must be supportable by the policy's asset share at the alteration date to avoid insurer loss.
- Boundary Conditions: Terms must be consistent with surrender, paid-up, and new policy values to prevent arbitrage.
- Stability: Small changes in benefits should not cause disproportionate changes in premiums or values.
- Simplicity: Methods should be easy to calculate and explain to policyholders.
- Incentives: Avoid creating incentives for lapse and re-entry or anti-selection.
- Profitability: The insurer should not make a loss; expected profit after alteration should be fair and consistent with new business.
- Expenses: Allowance for alteration and ongoing expenses must be made.
- Frequency: The impact of frequent alterations should be considered.
- Regulatory and professional guidance must be followed.

Boundary Conditions in Alterations

- Reduction in term to zero should match the cash surrender value.
- Reduction in sum assured should match paid-up calculations.
- Increase in benefits should be consistent with new policy pricing (may require underwriting and select mortality).
- Minor changes should result in minor value changes.

Methods of Calculating Alteration Terms

- Proportionate Paid-Up Value: Paid-up sum assured = Original sum assured × (premiums paid / premiums payable). Simple but not technically accurate.
- Equating Policy Values: Value before alteration = Prospective value after alteration (plus alteration expenses). More accurate and flexible, but complex and sensitive to basis selection.

Equating Policy Values Method

- Reserve before alteration equals the prospective reserve of the altered policy plus alteration expenses.
- Basis is usually the current pricing basis for new business at the date of alteration.
- Select mortality is used if new underwriting is required; otherwise, ultimate mortality applies.
- Profit is split into profit released at alteration (from the old basis) and profit emerging over the remaining term (from the new basis).

Alterations in Unit-Linked Contracts

- Paid-up status: Units remain, but penalties may be deducted to recoup initial expenses.

- Flexible adjustments: Premiums and benefits can often be changed; penalties may apply if future charges are reduced.
- Marketability may lead to some cross-subsidy if alteration volumes are low.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Principles and considerations for setting alteration terms** — Candidates must list and explain principles such as affordability, fairness, boundary conditions, stability, anti-selection, and expense recovery.
- **Suitability and implications of different alteration methods** — Discussing the pros and cons of methods like equating policy values versus proportionate paid-up, and their impact on profit, fairness, and policyholder expectations.
- **Boundary conditions and consistency** — Explaining how alteration terms should align with surrender, paid-up, and new policy values to prevent arbitrage and ensure fairness.
- **Profit emergence and basis selection** — Analyzing how profit is released or deferred under different bases and alteration methods, and the implications for the insurer and policyholder.
- **Practical calculation and adjustment for expenses** — Outlining how to adjust for initial, renewal, and alteration expenses in the calculation of new terms.
- **Paid-up versus surrender and alternatives** — Stating why insurers may prefer paid-up over surrender, and what alternatives exist for policyholders seeking reduced cover.
- **Unit-linked contract alterations** — Describing how alterations are handled in unit-linked business, including penalties and flexibility.
- **Impact of economic conditions and business context** — Considering how external factors (e.g., economic downturns) affect alteration rates and company strategy.

Common Pitfalls & Exam Technique

- Omitting key principles such as anti-selection, boundary conditions, or expense recovery in discussion questions.
- Failing to distinguish between the basis used before and after alteration, and not explaining profit emergence clearly.
- Assuming the proportionate paid-up method is always appropriate — it is simple but often unfair or inaccurate.
- Not checking for consistency with surrender and new business values, leading to potential arbitrage opportunities.
- Ignoring the need for underwriting or select mortality when benefits are increased.
- Overlooking the need to adjust for alteration expenses and to avoid double-charging initial expenses/commission.
- Not addressing the practicalities of explaining complex methods to policyholders.
- In calculation questions, not clearly stating all assumptions and not showing the full equation of value.

#18 Pricing health & care contracts LOW PRIORITY

2 exam questions | last tested 2023 | recency 1.16

What You Must Know

This topic covers the actuarial methods and considerations for pricing health and care insurance contracts, including income protection (IP), critical illness (CI), and long-term care insurance (LTCI). It is frequently examined due to the technical complexity of modeling morbidity risks, the need to select appropriate pricing models, and the practical challenges of data, assumptions, and experience monitoring.

Core Concepts

Income Protection (IP) Pricing Models

- **Possible IP States:** Basic states are healthy, sick, and dead; can be subdivided by risk factors or durations for more granular modeling.
- **Multi-State Model:** Tracks transitions between states (e.g., healthy to sick, sick to dead) using transition intensities; calculates EPV of benefits by integrating over time and transition probabilities.
- **Inception/Disabled Life Annuity Approach:** Uses claim inception rates (probability of becoming sick and remaining so through the deferred period) and disabled life annuity values to estimate benefit costs.
- **Deferred Period Adjustment:** Requires policyholder to remain sick for the deferred period before benefits are paid; modifies the EPV formula to include the probability of continuous sickness during the deferred period.
- **Integral Approximation:** In practice, integrals are approximated by summing over discrete time intervals, assuming average onset at mid-year, and using central claim inception rates.
- **Central Claim Inception Rates:** Represent expected claim inceptions over a year, relative to average lives exposed; used in summation formulas for EPV.
- **Annuity Rates Used:** Present value of annuities for in-claim lives (for claims reserves) and for healthy lives (for premium and gross premium valuation reserves).

Critical Illness (CI) Pricing

- **Risk Premium Derivation:** Sum incidence rates for each covered condition, adjust for overlaps/interactions, and multiply by survival probability (standalone) or combine with mortality (accelerated CI).
- **Data Sources:** Population data (adjusted for insured population), internal data (often insufficient), industry/reinsurer data (may lack consistency); issues include data scarcity, credibility, rapid obsolescence, and heterogeneity.

Long-Term Care Insurance (LTCI) Pricing

- **Multi-State Model:** Models transitions between multiple care states (e.g., healthy, home care, nursing home, dead); calculates EPV by summing over all care states and integrating transition probabilities.
- **Inception/Disability Annuity Model:** Sums expected costs across all care states, using incremental benefits for higher care levels and state-specific transition intensities; typically assumes no recovery from care states.

Model Selection: Advantages and Disadvantages

- **Multi-State Model:** Advantages include detailed insight, better sensitivity testing, and granular tracking; disadvantages are data scarcity, spurious accuracy, and operational complexity.
- **Inception/Disability Annuity Approach:** Advantages are practicality, avoidance of spurious accuracy, and common usage; disadvantages include limited sensitivity testing and less insight into rating/reserving robustness.

Examiner Angles — What They Actually Test

Based on past exam questions, the examiner consistently focuses on:

- **Describing and contrasting pricing models (IP):** Clearly explain both the inception/disabled life annuity and multi-state approaches, including their mechanics, assumptions, and differences.
- **Evaluating model change proposals:** Discuss practical, data, operational, and business factors when considering a switch from one pricing model to another.
- **Identifying causes of experience losses:** List and explain plausible reasons for adverse experience, such as data issues, incorrect assumptions, product definitions, underwriting, medical advances, and claims management.
- **Calculating and interpreting incidence rates (CI):** Apply formulas for claim incidence, including adjustments for survival periods and overlapping risks.
- **Discussing data sources and limitations (CI):** Critically assess the strengths and weaknesses of different data sources for morbidity assumptions.
- **Applying multi-state and annuity models (LTCI):** Demonstrate understanding of how to structure and calculate EPVs for LTCI using both multi-state and inception/disability annuity approaches.

Common Pitfalls & Exam Technique

- Confusing the inception/annuity approach with the multi-state model—be explicit about which model you are describing.
- Omitting the need for continuous sickness during the deferred period in IP pricing.
- Neglecting to discuss data credibility and operational feasibility when evaluating model changes.
- Failing to mention both advantages and disadvantages when comparing models.

- Not adjusting incidence rates for survival periods or overlapping risks in CI calculations.
- Overlooking the incremental nature of benefits in LTCI when moving between care states.
- In experience analysis questions, not covering a broad enough range of possible causes (e.g., data, product, process, environment).
- When asked for practical implementation, not mentioning the use of discrete approximations and central rates.